

A MINOR PROJECT REPORT
ON
POLYTECHNIC DIGITAL LIBRARY &
BOOK CIRCULATION MANAGEMENT SYSTEM

Submitted in partial fulfilment of the requirements
for the award of the diploma of

DIPLOMA IN INFORMATION TECHNOLOGY
(5th Semester)

SUBMITTED BY:

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SUBMITTED TO:

Department of Information Technology

MAHAMAYA POLYTECHNIC OF INFORMATION TECHNOLOGY
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DECLARATION

I, Prashant Kumar, a student of Diploma in Information Technology (5th Semester) at Mahamaya Polytechnic of Information Technology, Amroha, hereby declare that the minor project titled “Polytechnic Digital Library & Book Circulation Management System” is an original work carried out by me.

The matter embodied in this project report has not been submitted to any other University or Institution for the award of any degree or diploma. This project is entirely based on frontend web technologies (HTML, CSS, and JavaScript) and utilizes LocalStorage for data persistence, developed as part of my academic curriculum.

Name of Student: PRASHANT KUMAR

Enrollment No.: E23113835600018

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Date:

CERTIFICATE

This is to certify that the minor project report entitled “Polytechnic Digital Library & Book Circulation Management System” is a bonafide work carried out by Prashant Kumar (Enrollment No: E23113835600018), student of Diploma in Information Technology, 5th Semester at Mahamaya Polytechnic of Information Technology, Amroha.

This work has been completed in partial fulfilment of the requirements for the academic minor project evaluation. The project and report demonstrate a practical implementation of frontend software development skills.

.....
(Signature of HOD / Project Guide)

Department of Information Technology

ACKNOWLEDGMENT

The successful completion of any task would be incomplete without expressing gratitude to the people who made it possible. I would like to take this opportunity to express my profound gratitude and deep regards to the faculty members of the Department of Information Technology at Mahamaya Polytechnic of Information Technology, Amroha, for their exemplary guidance, monitoring, and constant encouragement throughout the course of this minor project.

I am deeply thankful to the Head of the Department (I.T) and my project guide for giving me the opportunity to work on this project and providing me with the necessary facilities and support.

Finally, I would like to thank my parents, friends, and peers for their continuous motivation and help during the development of this portal.

Prashant Kumar
(E23113835600018)

ABSTRACT

The 'Polytechnic Digital Library & Book Circulation Management System' is a lightweight, browser-based administrative portal designed specifically for managing book inventory and circulation in a polytechnic institute's departmental library.

Unlike traditional heavy library systems that require complex backend servers and databases, this minor project is developed entirely using modern client-side Web Technologies (HTML5, CSS3, and Vanilla JavaScript ES6+). The application logic heavily relies on DOM manipulation, Event Delegation, and JavaScript Date arithmetic, while ensuring complete data persistence across sessions using the browser's Web Storage API (LocalStorage).

The system features a dynamic dashboard, an inventory control module to add/edit/delete books, a robust circulation desk for issuing books to students, and an automated fine calculation mechanism that calculates overdue charges at ₹2 per day. Furthermore, it incorporates an official, printable Library Issue Slip generation feature for record-keeping.

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1. INTRODUCTION

1.1 Project Overview

The manual maintenance of library book records is a tedious, time-consuming, and error-prone process. To overcome this, the Polytechnic Digital Library System was conceptualized as a Minor Project. It acts as an interactive frontend interface that allows library administrators to track book stocks, student issues, and overdue returns seamlessly.

1.2 Objectives

- To digitize the departmental library circulation process without requiring server setup.
- To implement a responsive, user-friendly interface using standard web design principles.
- To automate the calculation of due dates (14-day loan policy) and late fines.
- To provide an easy-to-use search and filter mechanism for the library catalogue.

1.3 Scope of the Project

The current scope covers essential library management features suitable for a departmental library. It allows adding books, issuing them to students by capturing their Enrollment and branch details, tracking active records, and generating printable receipts. The data is secured locally on the operating device using LocalStorage, eliminating the need for database connectivity at this minor project stage.

2. TECHNOLOGY STACK

This project was purposefully built using core Web Technologies without any external dependencies, libraries, or frameworks (No React, No Bootstrap, No jQuery) to demonstrate a strong foundational grasp of web development.

2.1 HTML5 (HyperText Markup Language)

Semantic HTML5 is used to construct the page structure. Tags like `<header>`, `<main>`, `<section>`, `<article>`, and `<form>` were utilized to ensure the application is accessible and easily interpretable by browsers. Custom SVG and accessible form markup were strictly followed.

2.2 CSS3 (Cascading Style Sheets)

Modern CSS3 was employed for the interface layout.

- CSS Variables (`:root`) were used for consistent theming.
- Flexbox and CSS Grid were used for creating complex layouts like the stats dashboard and two-column forms.
- A dedicated '@media print' stylesheet was implemented to ensure the Library Issue Slip prints perfectly while hiding the rest of the UI.

2.3 JavaScript (ES6+)

Vanilla JavaScript drives the entire application logic.

- DOM Manipulation: Dynamic rendering of book tables, dropdown menus, and issue registries.
- Event Listeners: Handling form submissions, real-time search inputs, and modal triggers.
- Date Arithmetic: Automatically calculating the return due date by adding 14 days to the issue date, and calculating overdue days to generate fines.

2.4 LocalStorage API

Instead of a traditional database, the browser's Web Storage API (LocalStorage) was used to persist data via JSON serialization. State variables (`poly\_library\_books` and `poly\_library\_issued`) are continuously synced with LocalStorage so no data is lost upon page refresh.

Prasant MMIT Amroha

3. SYSTEM ARCHITECTURE & MODULES

The system is divided into functional modules that handle different administrative tasks.

3.1 Dashboard Module

Provides a live summary of the library. It reads the application state and dynamically displays the Total Books, Currently Issued Books, Overdue/Late Books, and Total Fine Collected in Indian Rupees.

3.2 Book Inventory Module (Collection Control)

Allows the librarian to add new books with details like Title, Author, Accession Code, Category, and Quantity. It includes a real-time search bar to filter the catalogue. The system also prevents the deletion of a book if it currently has an active issue record.

3.3 Book Circulation Module

This is the core functional area where books are issued. It captures the Student Name, Roll/Enrollment Number, Branch, and Semester. The form automatically assigns the current date as the issue date and calculates a strict 14-day due date.

3.4 Return & Fine Management

Displays a registry of all active circulations. It highlights overdue records in red. Upon returning a book, the system compares the return date against the due date. If overdue, it calculates a fine at ₹2 per overdue day and restores the book quantity back to the inventory.

3.5 Printable Issue Slip

Generates a formal modal dialog containing an issue receipt. It populates all student and book data into an official template, including placeholders for librarian and student signatures, which can be printed directly using the browser's print API.

4. PROJECT INTERFACE (SCREENSHOTS)

The following screenshots demonstrate the working interface of the Polytechnic Digital Library System. (Please insert the respective screenshots from your running project).

GitHub URL: <https://vts-space.github.io/Polytechnic-Digital-Library/>

The screenshot displays the Polytechnic Digital Library System interface. At the top, a dark blue header contains the library's name, department, and status. Below this, a light blue section titled 'LIBRARY ADMINISTRATION PORTAL' features a large heading 'Book circulation, kept simple.' and a brief description of the system's purpose. To the right, a 'CURRENT DESK DATE' is shown as 'Thursday, 24 September 2026'. Below these are four summary cards: 'Total Books in Library' (17), 'Books Currently Issued' (0), 'Overdue / Late Books' (0), and 'Total Fine Collected' (₹0). The main section, 'COLLECTION CONTROL', is titled 'Book inventory' and shows a 'Collection catalogue' table with 5 titles. On the left, a form for 'Add a new book' is visible, with fields for book title, author name, accession code, category, and available copies.

Polytechnic Digital Library
Department of Computer Science & Information Technology

Library Desk Online | 24 Sept 2026

LIBRARY ADMINISTRATION PORTAL

Book circulation, kept simple.

Manage your department collection, issue books to students, and keep every return accountable from one desk.

CURRENT DESK DATE
Thursday, 24 September 2026
Fine rate: ₹2 per overdue day

BK Total Books in Library
17
Available volumes

IS Books Currently Issued
0
Active circulation records

OD Overdue / Late Books
0
Needs follow-up today

₹ Total Fine Collected
₹0
From completed returns

COLLECTION CONTROL

Book inventory

5 titles

01 * Required

Add a new book

Book title *
e.g. Object-Oriented Programming

Author name *
e.g. Herbert Schildt

Accession / book code *
e.g. BK-105

Category *
Select category

Available copies *
e.g. 3

+ Add book to library

Cancel edit

Collection catalogue
Search by title or accession number.

Search books...

BOOK DETAILS	CATEGORY	STOCK	ACTION
Java Programming Herbert Schildt · BK-101	Computer Science	4	Edit Delete
Web Technology Uttam K. Roy · BK-102	IT	3	Edit Delete
Data Structures Seymour Lipschutz · BK-103	Computer Science	5	Edit Delete
Database Management Systems Raghu Ramakrishnan · BK-104	IT	2	Edit Delete
Engineering Mathematics - III B. S. Grewal · BK-105	Mathematics	3	Edit Delete

[*** SCREENSHOT 1: Book Inventory Form & Catalogue Table ***]

CIRCULATION DESK

Issue a book

14-day standard loan

02

Student details

Student full name *

Enrollment / roll number *

Branch *

Semester *

03

Loan details

Select available book *

Issue date *

Return due date

[→ Issue book](#)

[*** SCREENSHOT 2: Circulation Desk / Issue Form ***]

ACTIVE RECORDS

Issued books registry

2 active records

Current circulation

Returns automatically calculate the applicable fine.

₹2 / overdue day

STUDENT	BOOK TITLE	ISSUE DATE	DUE DATE	STATUS	ACTIONS
Vijay Prakash E23333835500012 - CSE - Semester 6	Java Programming BK-101	24 Sept 2026	08 Oct 2026	On Time	Print slip Return
Himanshu E23333838900002 - CSE - Semester 6	Data Structures BK-103	24 Sept 2026	08 Oct 2026	On Time	Print slip Return

[*** SCREENSHOT 3: Active Records & Issued books registry ***]



POLYTECHNIC INSTITUTE

Library Issue Slip

Department of Computer Science & Information Technology

Receipt No. **LIB-1790270029866**

Issued on **24 Sept 2026**

STUDENT NAME

Himanshu

ENROLLMENT / ROLL NO.

E23333838900002

BRANCH & SEMESTER

CSE - Semester 6

ACCESSION NO.

BK-103

BOOK ISSUED

Data Structures

Issue date: **24 Sept 2026**

Return due: **08 Oct 2026**

Library policy: Return the book on or before the due date. A late fine of ₹2 per day will be charged after the due date.

Librarian signature

Student signature

 **Print / Save PDF**

[*** SCREENSHOT 4: Active Records & Printable Issue Slip Modal ***]

5. IMPLEMENTATION & SOURCE CODE

The complete application is built across three primary files. Below are the crucial segments of the source code that drive the application.

5.1 HTML Structure (index.html)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Polytechnic Digital Library and Book Circulation Management System">
  <link rel="icon" type="image/png" href="learning.png">
  <title>Polytechnic Digital Library | Circulation Desk</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
  <header class="site-header">
    <div class="header-inner">
      <a class="brand" href="#dashboard" aria-label="Polytechnic Digital Library home">
        
        <span>
          <strong>Polytechnic Digital Library</strong>
          <small>Department of Computer Science & Information Technology</small>
        </span>
      </a>
      <div class="header-meta">
        <span class="status-dot" aria-hidden="true"></span>
        <span>Library Desk Online</span>
        <time id="todayLabel"></time>
      </div>
    </div>
  </header>

  <main class="page-shell">
    <section class="hero" id="dashboard" aria-labelledby="pageTitle">
      <div>
        <p class="eyebrow">LIBRARY ADMINISTRATION PORTAL</p>
        <h1 id="pageTitle">Book circulation, kept simple.</h1>
        <p class="hero-copy">Manage your department collection, issue books to students, and keep every return accountable from one desk.</p>
      </div>
      <div class="hero-note">
        <span class="hero-note-label">Current desk date</span>
        <strong id="heroDate"></strong>
        <span>Fine rate: ₹2 per overdue day</span>
      </div>
    </section>

    <section class="stats-grid" aria-label="Library summary">
```

```

        <article class="stat-card stat-blue">
            <div class="stat-icon" aria-hidden="true">BK</div>
            <div><span class="stat-label">Total Books in Library</span><strong
id="totalBooks">0</strong><span
                class="stat-detail">Available volumes</span></div>
        </article>
        <article class="stat-card stat-amber">
            <div class="stat-icon" aria-hidden="true">IS</div>
            <div><span class="stat-label">Books Currently Issued</span><strong
id="issuedBooks">0</strong><span
                class="stat-detail">Active circulation records</span></div>
        </article>
        <article class="stat-card stat-red">
            <div class="stat-icon" aria-hidden="true">OD</div>
            <div><span class="stat-label">Overdue / Late Books</span><strong
id="overdueBooks">0</strong><span
                class="stat-detail">Needs follow-up today</span></div>
        </article>
        <article class="stat-card stat-green">
            <div class="stat-icon" aria-hidden="true">₹</div>
            <div><span class="stat-label">Total Fine Collected</span><strong
id="totalFine">₹0</strong><span
                class="stat-detail">From completed returns</span></div>
        </article>
    </section>

    <div id="notice" class="notice" role="status" aria-live="polite"></div>

    <section class="content-section" id="inventory" aria-
labelledby="inventoryTitle">
        <div class="section-heading">
            <div>
                <p class="section-kicker">COLLECTION CONTROL</p>
                <h2 id="inventoryTitle">Book inventory</h2>
            </div>
            <span class="record-count" id="bookCount">0 titles</span>
        </div>
        <div class="module-grid">
            <form class="panel form-panel" id="bookForm">
                <div class="panel-heading">
                    <div><span class="panel-index">01</span>
                        <h3 id="bookFormTitle">Add a new book</h3>
                    </div><span class="required-note">* Required</span>
                </div>
                <label for="bookTitle">Book title <span>*</span></label>
                <input id="bookTitle" name="title" type="text" placeholder="e.g.
Object-Oriented Programming"
                    required maxlength="120">
                <label for="bookAuthor">Author name <span>*</span></label>
                <input id="bookAuthor" name="author" type="text" placeholder="e.g.
Herbert Schildt" required
                    maxlength="100">
                <label for="bookCode">Accession / book code <span>*</span></label>
                <input id="bookCode" name="code" type="text" placeholder="e.g. BK-
105" pattern="[A-Za-z0-9-]+"
                    required maxlength="20">
                <div class="two-col">
                    <div><label for="bookCategory">Category
<span>*</span></label><select id="bookCategory"
                        name="category" required>
                            <option value="">Select category</option>

```

```

        <option>Computer Science</option>
        <option>IT</option>
        <option>Electronics</option>
        <option>Mathematics</option>
        <option>General</option>
    </select></div>
    <div><label for="bookQuantity">Available copies
<span>*</span></label><input id="bookQuantity"
        name="quantity" type="number" min="1" step="1"
placeholder="e.g. 3" required></div>
    </div>
    <button class="button button-primary" id="bookSubmitButton"
type="submit"><span
        aria-hidden="true">+</span> Add book to
library</button><button
        class="button button-secondary" id="cancelBookEdit"
type="button" hidden>Cancel edit</button>
    </form>

    <div class="panel table-panel">
        <div class="table-toolbar">
            <div>
                <h3>Collection catalogue</h3>
                <p>Search by title or accession number.</p>
            </div><label class="search-box" for="bookSearch"><span aria-
hidden="true">ð</span><input
                id="bookSearch" type="search" placeholder="Search
books..."
                aria-label="Search books"></label>
            </div>
            <div class="table-wrap">
                <table>
                    <thead>
                        <tr>
                            <th>Book details</th>
                            <th>Category</th>
                            <th>Stock</th>
                            <th>Action</th>
                        </tr>
                    </thead>
                    <tbody id="booksTableBody"></tbody>
                </table>
            </div>
            <div class="empty-state" id="booksEmpty" hidden><span
class="empty-icon"
                aria-hidden="true">BK</span><strong>No books match your
search</strong>
                <p>Try another title or accession number.</p>
            </div>
        </div>
    </div>
</section>

    <section class="content-section circulation-section" id="circulation" aria-
labelledby="circulationTitle">
        <div class="section-heading">
            <div>
                <p class="section-kicker">CIRCULATION DESK</p>
                <h2 id="circulationTitle">Issue a book</h2>
            </div><span class="desk-label">14-day standard loan</span>
        </div>

```



```

        <div>
            <p class="section-kicker">ACTIVE RECORDS</p>
            <h2 id="issuedTitle">Issued books registry</h2>
        </div><span class="record-count" id="issuedCount">0 active
records</span>
    </div>
    <div class="panel table-panel registry-panel">
        <div class="table-toolbar">
            <div>
                <h3>Current circulation</h3>
                <p>Returns automatically calculate the applicable fine.</p>
            </div><span class="fine-key"><span></span> ₹2 / overdue day</span>
        </div>
        <div class="table-wrap">
            <table>
                <thead>
                    <tr>
                        <th>Student</th>
                        <th>Book title</th>
                        <th>Issue date</th>
                        <th>Due date</th>
                        <th>Status</th>
                        <th>Actions</th>
                    </tr>
                </thead>
                <tbody id="issuedTableBody"></tbody>
            </table>
        </div>
        <div class="empty-state" id="issuedEmpty" hidden><span class="empty-
icon"
            aria-hidden="true">OK</span><strong>No active issue
records</strong>
            <p>Issued books will appear here for easy tracking.</p>
        </div>
    </div>
</section>
</main>

<footer class="site-footer">
    <div><strong>Polytechnic Digital Library</strong><span>Academic resource
management portal</span></div>
    <span>Data is stored locally on this device.</span><span>Made with VTS
SPACE.</span>
</footer>

<div class="modal-backdrop" id="receiptModal" hidden>
    <section class="receipt-modal" role="dialog" aria-modal="true" aria-
labelledby="receiptTitle"><button
        class="modal-close" id="closeReceipt" type="button" aria-label="Close
receipt">x</button>
        <div class="receipt-paper" id="receiptPaper">
            <div class="receipt-head"><span class="receipt-emblem">DL</span>
            <div>
                <p>POLYTECHNIC INSTITUTE</p>
                <h2 id="receiptTitle">Library Issue Slip</h2><span>Department
of Computer Science &amp;
                    Information Technology</span>
            </div>
        </div>
    </div>
    <div class="receipt-rule"></div>

```

```

        <div class="receipt-meta"><span>Receipt No. <strong
id="receiptNumber"></strong></span><span>Issued on
        <strong id="receiptIssuedOn"></strong></span></div>
        <div class="receipt-grid">
            <div><span>Student name</span><strong
id="receiptStudent"></strong></div>
            <div><span>Enrollment / roll no.</span><strong
id="receiptRoll"></strong></div>
            <div><span>Branch & semester</span><strong
id="receiptBranch"></strong></div>
            <div><span>Accession no.</span><strong
id="receiptCode"></strong></div>
        </div>
        <div class="receipt-book"><span>BOOK ISSUED</span><strong
id="receiptBook"></strong>
            <div><span>Issue date: <b
id="receiptIssueDate"></b></span><span>Return due: <b
id="receiptDueDate"></b></span></div>
        </div>
        <p class="receipt-policy"><strong>Library policy:</strong> Return the
book on or before the due date. A
        late fine of ₹2 per day will be charged after the due date.</p>
        <div class="signature-row">
            <div><span></span><small>Librarian signature</small></div>
            <div><span></span><small>Student signature</small></div>
        </div>
        </div><button class="button button-primary print-button" id="printReceipt"
type="button"><span
        aria-hidden="true">🖨️</span> Print / Save PDF</button>
    </section>
</div>

<div class="app-dialog-backdrop" id="appDialog" hidden>
    <section class="app-dialog" role="dialog" aria-modal="true" aria-
labelledby="dialogTitle"
        aria-describedby="dialogMessage">
        <div class="dialog-mark" aria-hidden="true">!</div>
        <div class="dialog-content">
            <p class="dialog-kicker">LIBRARY DESK</p>
            <h2 id="dialogTitle">Confirm action</h2>
            <p id="dialogMessage"></p>
        </div>
        <div class="dialog-actions"><button class="dialog-button dialog-cancel"
id="dialogCancel"
            type="button">Cancel</button><button class="dialog-button dialog-
confirm" id="dialogConfirm"
            type="button">Confirm</button></div>
    </section>
</div>

<script src="script.js"></script>
</body>

</html>

```

5.2 CSS Styling (style.css)

```

:root {
  --ink: #172334;
  --muted: #697789;
  --line: #dce3ea;
  --paper: #ffffff;
  --canvas: #f4f7fa;
  --navy: #173d64;
  --blue: #1769aa;
  --blue-soft: #e8f2fb;
  --green: #16805d;
  --green-soft: #e8f6f0;
  --red: #c74444;
  --red-soft: #fff0ef;
  --amber: #b26b13;
  --amber-soft: #fff5df;
  --shadow: 0 14px 35px rgba(28, 54, 78,
.08);
  --radius: 12px;
  font-family: "Segoe UI", Tahoma, sans-
serif;
  color: var(--ink);
  background: var(--canvas);
}

* {
  box-sizing: border-box;
}

* {
  scrollbar-width: none;
  -ms-overflow-style: none;
}

*::-webkit-scrollbar {
  width: 0;
  height: 0;
}

html {
  scroll-behavior: smooth;
}

body {
  margin: 0;
  background: var(--canvas);
  color: var(--ink);
  font-size: 14px;
  line-height: 1.5;
}

button,
input,
select {
  font: inherit;
}

button {
  cursor: pointer;
}

.site-header {
  background: var(--navy);
  color: #fff;
  border-bottom: 3px solid #e4ad52;
}

.header-inner,
.page-shell,
.site-footer {
  width: min(1240px, calc(100% - 48px));
  margin: auto;
}

.header-inner {
  min-height: 76px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 24px;
}

.brand {
  color: inherit;
  text-decoration: none;
  display: flex;
  gap: 12px;
  align-items: center;
}

.brand-mark,
.receipt-emblem {
  display: grid;
  place-items: center;
  width: 42px;
  height: 42px;
  border-radius: 8px;
}

.brand-mark {
  object-fit: cover;
}

.receipt-emblem {
  background: #e4ad52;
  color: var(--navy);
  font-weight: 800;
  letter-spacing: .04em;
}

.brand strong {
  display: block;
  font-family: Georgia, serif;
  font-size: 17px;
  letter-spacing: .01em;
}

.brand small {
  display: block;
}

```

```

margin-top: 2px;
color: #cbd9e7;
font-size: 11px;
}

.header-meta {
  display: flex;
  align-items: center;
  gap: 10px;
  color: #d4e0eb;
  font-size: 12px;
}

.header-meta time {
  margin-left: 12px;
  padding-left: 16px;
  border-left: 1px solid #52718e;
}

.status-dot {
  width: 8px;
  height: 8px;
  border-radius: 50%;
  background: #6bd7a8;
  box-shadow: 0 0 0 4px rgba(107, 215,
168, .15);
}

.page-shell {
  padding: 46px 0 76px;
}

.hero {
  display: flex;
  justify-content: space-between;
  align-items: end;
  gap: 30px;
  margin-bottom: 32px;
}

.eyebrow,
.section-kicker {
  color: var(--blue);
  font-size: 11px;
  font-weight: 800;
  letter-spacing: .14em;
  margin: 0 0 7px;
}

h1,
h2,
h3,
p {
  margin-top: 0;
}

h1 {
  max-width: 600px;
  margin-bottom: 9px;
  font-family: Georgia, serif;
  font-size: clamp(34px, 5vw, 54px);
  font-weight: 600;
  line-height: 1.05;
  letter-spacing: -.02em;
}

.hero-copy {
  max-width: 560px;
  margin: 0;
  color: var(--muted);
  font-size: 16px;
}

.hero-note {
  min-width: 210px;
  border-left: 3px solid #e4ad52;
  padding: 6px 0 6px 18px;
  color: var(--muted);
}

.hero-note-label {
  display: block;
  font-size: 11px;
  font-weight: 700;
  letter-spacing: .1em;
  text-transform: uppercase;
}

.hero-note strong {
  display: block;
  margin: 3px 0 5px;
  color: var(--navy);
  font-family: Georgia, serif;
  font-size: 20px;
}

.hero-note span:last-child {
  font-size: 12px;
}

.stats-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 16px;
  margin-bottom: 48px;
}

.stat-card {
  display: flex;
  align-items: flex-start;
  gap: 14px;
  min-height: 124px;
  padding: 22px 18px;
  background: var(--paper);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
}

```

```

        font-weight: 600;
        font-size: 13px;
        transition: all .2s ease;
    }

.stat-icon {
    flex: 0 0 36px;
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    border-radius: 8px;
    font-size: 11px;
    font-weight: 800;
}

.stat-blue .stat-icon {
    background: var(--blue-soft);
    color: var(--blue);
}

.stat-amber .stat-icon {
    background: var(--amber-soft);
    color: var(--amber);
}

.stat-red .stat-icon {
    background: var(--red-soft);
    color: var(--red);
}

.stat-green .stat-icon {
    background: var(--green-soft);
    color: var(--green);
}

.stat-label,
.stat-detail {
    display: block;
    color: var(--muted);
    font-size: 12px;
}

.stat-card strong {
    display: block;
    margin: 5px 0 1px;
    color: var(--navy);
    font-family: Georgia, serif;
    font-size: 28px;
    line-height: 1;
}

.stat-detail {
    color: #98a3af;
    font-size: 11px;
}

.notice {
    min-height: 0;
    margin: -28px 0 28px;
    padding: 0;
    border-radius: 8px;
    color: var(--green);

    notice:not(:empty) {
        padding: 11px 15px;
        background: var(--green-soft);
        border: 1px solid #bde6d5;
    }

    notice.error {
        color: var(--red);
        background: var(--red-soft);
        border-color: #f2c2bf;
    }

    .content-section {
        margin-top: 52px;
        scroll-margin-top: 24px;
    }

    .section-heading {
        display: flex;
        align-items: end;
        justify-content: space-between;
        gap: 20px;
        margin-bottom: 18px;
    }

    .section-kicker {
        margin-bottom: 4px;
        font-size: 10px;
    }

    h2 {
        margin-bottom: 0;
        color: var(--navy);
        font-family: Georgia, serif;
        font-size: 30px;
        font-weight: 600;
    }

    h3 {
        margin: 0;
        color: var(--navy);
        font-size: 16px;
    }

    .record-count,
    .desk-label {
        color: var(--muted);
        font-size: 12px;
        font-weight: 600;
    }

    .desk-label {
        padding: 7px 11px;
        color: var(--blue);
    }

```

```

background: var(--blue-soft);
border-radius: 5px;
}

.module-grid {
display: grid;
grid-template-columns: minmax(260px,
320px) 1fr;
gap: 18px;
align-items: start;
}

.panel {
background: var(--paper);
border: 1px solid var(--line);
border-radius: var(--radius);
box-shadow: var(--shadow);
}

.form-panel {
padding: 24px;
}

.panel-heading {
display: flex;
align-items: flex-start;
justify-content: space-between;
margin-bottom: 22px;
}

.panel-index {
display: block;
margin-bottom: 5px;
color: #c28a31;
font-size: 10px;
font-weight: 800;
letter-spacing: .14em;
}

.required-note {
color: #99a4b0;
font-size: 11px;
}

label {
display: block;
margin: 15px 0 6px;
color: #516173;
font-size: 12px;
font-weight: 700;
}

label span {
color: var(--red);
}

input,
select {
width: 100%;

```

```

height: 42px;
padding: 0 12px;
color: var(--ink);
background: #fbfcfd;
border: 1px solid #cbd6e0;
border-radius: 6px;
outline: none;
transition: border .15s, box-shadow
.15s;
}

select {
appearance: none;
padding-right: 42px;
cursor: pointer;
background-color: #fbfcfd;
background-image: linear-gradient(45deg,
transparent 50%, var(--navy) 50%), linear-
gradient(135deg, var(--navy) 50%,
transparent 50%);
background-position: calc(100% - 17px)
18px, calc(100% - 12px) 18px;
background-repeat: no-repeat;
background-size: 5px 5px, 5px 5px;
}

select:hover {
border-color: #9fb2c2;
background-color: #f6f9fb;
}

select:focus {
border-color: var(--blue);
box-shadow: 0 0 0 3px rgba(23, 105, 170,
.12);
background-color: #fff;
}

select option {
color: var(--ink);
background: #fff;
}

select option:first-child {
color: var(--muted);
}

input:focus {
border-color: var(--blue);
box-shadow: 0 0 0 3px rgba(23, 105, 170,
.12);
}

.custom-select {
position: relative;
width: 100%;
}

.custom-select .native-select-hidden {

```

```

    position: absolute;
    width: 1px;
    height: 1px;
    padding: 0;
    opacity: 0;
    pointer-events: none;
}

.custom-select-trigger {
    display: flex;
    align-items: center;
    justify-content: space-between;
    width: 100%;
    height: 42px;
    padding: 0 12px;
    color: var(--ink);
    background: #fbfcfd;
    border: 1px solid #cbd6e0;
    border-radius: 6px;
    text-align: left;
    transition: border .15s, box-shadow
.15s, background .15s;
}

.custom-select-trigger:hover,
.custom-select.is-open .custom-select-
trigger {
    border-color: var(--blue);
    background: #fff;
}

.custom-select-trigger:focus-visible {
    outline: none;
    border-color: var(--blue);
    box-shadow: 0 0 0 3px rgba(23, 105, 170,
.12);
}

.custom-select-trigger .select-placeholder {
    color: var(--muted);
}

.custom-select-trigger .select-chevron {
    width: 8px;
    height: 8px;
    margin-right: 2px;
    border-right: 2px solid var(--navy);
    border-bottom: 2px solid var(--navy);
    transform: translateY(-2px)
rotate(45deg);
    transition: transform .18s ease;
}

.custom-select.is-open .select-chevron {
    transform: translateY(2px)
rotate(225deg);
}

.custom-select-menu {
    position: absolute;
    z-index: 5;
    top: calc(100% + 7px);
    left: 0;
    right: 0;
    display: none;
    max-height: 230px;
    overflow-y: auto;
    padding: 6px;
    background: #fff;
    border: 1px solid #cbd6e0;
    border-radius: 8px;
    box-shadow: 0 14px 28px rgba(23, 61,
100, .16);
}

.custom-select.is-open .custom-select-menu {
    display: block;
    animation: select-menu-in .16s ease-out;
}

.custom-select-option {
    display: flex;
    align-items: center;
    width: 100%;
    min-height: 37px;
    padding: 8px 10px;
    color: var(--ink);
    background: transparent;
    border: 0;
    border-radius: 5px;
    text-align: left;
    font-size: 13px;
    cursor: pointer;
}

.custom-select-option:hover,
.custom-select-option:focus-visible {
    color: var(--navy);
    background: var(--blue-soft);
    outline: none;
}

.custom-select-option.is-selected {
    color: var(--blue);
    background: #f0f6fb;
    font-weight: 700;
}

.custom-select-option.is-placeholder {
    color: var(--muted);
    font-weight: 400;
}

@keyframes select-menu-in {
    from {
        opacity: 0;
        transform: translateY(-4px);
    }
}

```

```

    to {
      opacity: 1;
      transform: translateY(0);
    }
  }

input::placeholder {
  color: #a5afb9;
}

input[readonly] {
  background: #f1f5f8;
  color: var(--muted);
}

.two-col {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 12px;
}

.button {
  display: inline-flex;
  justify-content: center;
  align-items: center;
  gap: 8px;
  min-height: 42px;
  padding: 0 16px;
  border: 0;
  border-radius: 6px;
  font-weight: 700;
  transition: transform .15s, background
.15s;
}

.button:hover {
  transform: translateY(-1px);
}

.button-primary {
  width: 100%;
  margin-top: 22px;
  color: #fff;
  background: var(--blue);
  box-shadow: 0 5px 12px rgba(23, 105,
170, .18);
}

.button-primary:hover {
  background: #12588e;
}

.button-secondary {
  width: 100%;
  margin-top: 9px;
  color: var(--navy);
  background: #eef3f7;
  border: 1px solid #d6e0e8;
}

```

```

.button-secondary:hover {
  background: #e3ebf1;
}

```

```

.table-panel {
  min-width: 0;
  overflow: hidden;
}

```

```

.table-toolbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 18px;
  padding: 21px 22px 16px;
}

```

```

.table-toolbar p {
  margin: 3px 0 0;
  color: var(--muted);
  font-size: 12px;
}

```

```

.search-box {
  position: relative;
  display: flex;
  align-items: center;
  width: min(230px, 45%);
  margin: 0;
}

```

```

.search-box span {
  position: absolute;
  left: 12px;
  color: var(--muted);
  font-size: 22px;
  line-height: 1;
}

```

```

.search-box input {
  height: 38px;
  padding-left: 34px;
}

```

```

.table-wrap {
  overflow-x: auto;
}

```

```

table {
  width: 100%;
  border-collapse: collapse;
  white-space: nowrap;
}

```

```

th {
  padding: 10px 22px;
  color: #83909d;
  background: #f8fafc;
}

```

```

    font-size: 10px;
    font-weight: 800;
    letter-spacing: .1em;
    text-align: left;
    text-transform: uppercase;
}

td {
    padding: 15px 22px;
    border-top: 1px solid #edf1f4;
    vertical-align: middle;
    font-size: 13px;
}

.book-detail strong,
.student-detail strong {
    display: block;
    color: var(--navy);
    font-weight: 700;
}

.book-detail span,
.student-detail span,
td small {
    color: var(--muted);
    font-size: 11px;
}

.stock {
    display: inline-block;
    min-width: 35px;
    padding: 4px 8px;
    color: var(--green);
    background: var(--green-soft);
    border-radius: 5px;
    text-align: center;
    font-weight: 700;
}

.stock.empty {
    color: var(--red);
    background: var(--red-soft);
}

.text-button {
    padding: 4px 0;
    color: var(--red);
    background: transparent;
    border: 0;
    font-size: 12px;
    font-weight: 700;
}

.text-button.edit {
    color: var(--blue);
}

.text-button:hover {
    text-decoration: underline;
}

}

.issue-form {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 0;
    overflow: visible;
}

.form-block {
    padding: 26px 28px;
}

.form-block+.form-block {
    border-left: 1px solid var(--line);
}

.form-block h3 {
    margin-bottom: 18px;
}

.form-block .button {
    margin-top: 22px;
}

.registry-panel {
    overflow: hidden;
}

.fine-key {
    color: var(--muted);
    font-size: 11px;
}

.fine-key span {
    display: inline-block;
    width: 7px;
    height: 7px;
    margin-right: 5px;
    background: var(--red);
    border-radius: 50%;
}

.status-badge {
    display: inline-block;
    padding: 5px 8px;
    border-radius: 5px;
    font-size: 11px;
    font-weight: 700;
}

.status-badge.on-time {
    color: var(--green);
    background: var(--green-soft);
}

.status-badge.overdue {
    color: var(--red);
    background: var(--red-soft);
}

```

```

}
margin: 4px 0 0;
font-size: 12px;

.action-group {
    display: flex;
    align-items: center;
    gap: 13px;
}

.print-link {
    color: var(--blue);
    background: transparent;
    border: 0;
    font-size: 12px;
    font-weight: 700;
}

.print-link:hover {
    text-decoration: underline;
}

.return-button {
    padding: 7px 10px;
    color: #fff;
    background: var(--navy);
    border: 0;
    border-radius: 5px;
    font-size: 11px;
    font-weight: 700;
}

.return-button:hover {
    background: #285b87;
}

.empty-state {
    padding: 42px 20px;
    color: var(--muted);
    text-align: center;
}

.empty-icon {
    display: grid;
    place-items: center;
    width: 40px;
    height: 40px;
    margin: 0 auto 10px;
    color: var(--blue);
    background: var(--blue-soft);
    border-radius: 8px;
    font-size: 11px;
    font-weight: 800;
}

.empty-state strong {
    display: block;
    color: var(--navy);
}

.empty-state p {
margin: 4px 0 0;
font-size: 12px;

.site-footer {
    display: flex;
    justify-content: space-between;
    gap: 20px;
    padding: 26px 0 34px;
    color: #8996a3;
    border-top: 1px solid var(--line);
    font-size: 11px;
}

.site-footer strong,
.site-footer span {
    display: block;
}

.site-footer strong {
    color: var(--navy);
    font-size: 12px;
}

.modal-backdrop {
    position: fixed;
    inset: 0;
    z-index: 10;
    display: grid;
    place-items: center;
    padding: 24px;
    background: rgba(17, 35, 54, .62);
}

.modal-backdrop[hidden] {
    display: none;
}

.receipt-modal {
    position: relative;
    width: min(580px, 100%);
    max-height: calc(100vh - 48px);
    overflow: auto;
    padding: 25px;
    background: #eef2f5;
    border-radius: 12px;
    box-shadow: 0 24px 60px rgba(0, 0, 0,
.25);
}

.modal-close {
    position: absolute;
    top: 8px;
    right: 12px;
    z-index: 1;
    width: 32px;
    height: 32px;
    color: var(--muted);
    background: transparent;
}

```

```

border: 0;
font-size: 25px;
line-height: 1;
}

.receipt-paper {
padding: 28px;
background: #fff;
box-shadow: 0 5px 15px rgba(26, 48, 68,
.1);
}

.receipt-head {
display: flex;
align-items: center;
gap: 12px;
}

.receipt-emblem {
flex: 0 0 42px;
}

.receipt-head p {
margin: 0;
color: var(--blue);
font-size: 10px;
font-weight: 800;
letter-spacing: .1em;
}

.receipt-head h2 {
margin: 1px 0;
font-size: 24px;
}

.receipt-head span:last-child {
color: var(--muted);
font-size: 10px;
}

.receipt-rule {
height: 2px;
margin: 22px 0 12px;
background: var(--navy);
}

.receipt-meta {
display: flex;
justify-content: space-between;
color: var(--muted);
font-size: 10px;
}

.receipt-meta strong {
color: var(--ink);
}

.receipt-grid {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 16px 24px;
margin: 25px 0;
}

.receipt-grid span,
.receipt-book>span {
display: block;
color: #8996a3;
font-size: 10px;
font-weight: 700;
letter-spacing: .08em;
text-transform: uppercase;
}

.receipt-grid strong {
display: block;
margin-top: 3px;
font-size: 13px;
}

.receipt-book {
padding: 17px;
background: #f3f7fa;
border-left: 3px solid #e4ad52;
}

.receipt-book>strong {
display: block;
margin: 5px 0 12px;
color: var(--navy);
font-family: Georgia, serif;
font-size: 18px;
}

.receipt-book div {
display: flex;
justify-content: space-between;
color: var(--muted);
font-size: 11px;
}

.receipt-policy {
margin: 22px 0 35px;
color: var(--muted);
font-size: 11px;
}

.receipt-policy strong {
color: var(--ink);
}

.signature-row {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 40px;
}

.signature-row span {

```

```

    display: block;
    height: 28px;
    border-bottom: 1px solid var(--ink);
}

.signature-row small {
    display: block;
    margin-top: 5px;
    color: var(--muted);
    font-size: 10px;
}

.print-button {
    width: 100%;
    margin-top: 18px;
}

.app-dialog-backdrop {
    position: fixed;
    inset: 0;
    z-index: 20;
    display: grid;
    place-items: center;
    padding: 20px;
    background: rgba(12, 29, 46, .58);
    backdrop-filter: blur(3px);
}

.app-dialog-backdrop[hidden] {
    display: none;
}

.app-dialog {
    width: min(420px, 100%);
    padding: 25px;
    background: #fff;
    border: 1px solid #d9e3eb;
    border-radius: 14px;
    box-shadow: 0 24px 70px rgba(12, 35, 57, .28);
    animation: dialog-in .18s ease-out;
}

.dialog-mark {
    display: grid;
    place-items: center;
    width: 38px;
    height: 38px;
    margin-bottom: 16px;
    color: #a66312;
    background: var(--amber-soft);
    border: 1px solid #f1d59e;
    border-radius: 50%;
    font-weight: 800;
}

.dialog-kicker {
    margin: 0 0 5px;
    color: var(--blue);
    font-size: 10px;
    font-weight: 800;
    letter-spacing: .14em;
}

.app-dialog h2 {
    margin-bottom: 8px;
    font-size: 25px;
}

.app-dialog p {
    margin-bottom: 0;
    color: var(--muted);
    font-size: 14px;
    line-height: 1.55;
}

.dialog-actions {
    display: flex;
    justify-content: flex-end;
    gap: 9px;
    margin-top: 25px;
}

.dialog-button {
    min-height: 39px;
    padding: 0 15px;
    border-radius: 6px;
    font-weight: 700;
}

.dialog-cancel {
    color: var(--muted);
    background: #f1f4f7;
    border: 1px solid #d6dfe7;
}

.dialog-cancel:hover {
    color: var(--ink);
    background: #e8eef3;
}

.dialog-confirm {
    color: #fff;
    background: var(--blue);
    border: 1px solid var(--blue);
}

.dialog-confirm:hover {
    background: #12588e;
}

@keyframes dialog-in {
    from {
        opacity: 0;
        transform: translateY(8px)
    }
    scale(.98);
}

```

```

    to {
      opacity: 1;
      transform: translateY(0) scale(1);
    }
  }

  @media (max-width: 900px) {
    .stats-grid {
      grid-template-columns: repeat(2,
1fr);
    }

    .module-grid {
      grid-template-columns: 1fr;
    }
  }

  @media (max-width: 650px) {

    .header-inner,
    .page-shell,
    .site-footer {
      width: min(100% - 28px, 1240px);
    }

    .header-inner {
      padding: 13px 0;
      align-items: flex-start;
    }

    .header-meta {
      display: none;
    }

    .hero {
      display: block;
    }

    .hero-note {
      margin-top: 24px;
    }

    .stats-grid {
      gap: 10px;
    }

    .stat-card {
      min-height: 112px;
      padding: 15px 12px;
    }

    .stat-card strong {
      font-size: 24px;
    }

    .stat-label {
      font-size: 11px;
    }
  }

```

```

.section-heading {
  align-items: flex-start;
}

.section-heading h2 {
  font-size: 26px;
}

.table-toolbar {
  display: block;
}

.search-box {
  width: 100%;
  margin-top: 14px;
}

.issue-form {
  display: block;
}

.form-block+.form-block {
  border-top: 1px solid var(--line);
  border-left: 0;
}

.site-footer {
  display: block;
}

.site-footer>span {
  margin-top: 13px;
}

.receipt-modal {
  padding: 10px;
}

.receipt-paper {
  padding: 20px 16px;
}
}

@media print {
  body>*:not(.modal-backdrop) {
    display: none !important;
  }

  .modal-backdrop {
    position: static;
    display: block;
    padding: 0;
    background: none;
  }

  .receipt-modal {
    width: 100%;
    max-height: none;
    padding: 0;
  }
}

```

```

        background: none;
        box-shadow: none;
    }

    .receipt-paper {
        box-shadow: none;
    }

    .modal-close,
    .print-button {
        display: none !important;
    }

```

5.3 JavaScript Logic (script.js)

```
/*
 * Polytechnic Digital Library
 * Vanilla JavaScript state, DOM rendering, and LocalStorage persistence.
 */

const BOOKS_KEY = "poly_library_books";
const ISSUED_KEY = "poly_library_issued";
const FINE_PER_DAY = 2;
const LOAN_DAYS = 14;

const sampleBooks = [
  { id: "book-101", title: "Java Programming", author: "Herbert Schildt", code: "BK-101",
    category: "Computer Science", quantity: 4 },
  { id: "book-102", title: "Web Technology", author: "Uttam K. Roy", code: "BK-102",
    category: "IT", quantity: 3 },
  { id: "book-103", title: "Data Structures", author: "Seymour Lipschutz", code: "BK-103",
    category: "Computer Science", quantity: 5 },
  { id: "book-104", title: "Database Management Systems", author: "Raghu Ramakrishnan", code:
    "BK-104", category: "IT", quantity: 2 },
  { id: "book-105", title: "Engineering Mathematics - III", author: "B. S. Grewal", code:
    "BK-105", category: "Mathematics", quantity: 3 }
];

let books = readStorage(BOOKS_KEY, sampleBooks);
let issuedRecords = readStorage(ISSUED_KEY, []);
let editingBookId = null;

const elements = {
  todayLabel: document.querySelector("#todayLabel"),
  heroDate: document.querySelector("#heroDate"),
  notice: document.querySelector("#notice"),
  bookForm: document.querySelector("#bookForm"),
  bookFormTitle: document.querySelector("#bookFormTitle"),
  bookSubmitButton: document.querySelector("#bookSubmitButton"),
  cancelBookEdit: document.querySelector("#cancelBookEdit"),
  bookSearch: document.querySelector("#bookSearch"),
  booksBody: document.querySelector("#booksTableBody"),
  booksEmpty: document.querySelector("#booksEmpty"),
  bookCount: document.querySelector("#bookCount"),
  issueForm: document.querySelector("#issueForm"),
  issueBook: document.querySelector("#issueBook"),
  issueDate: document.querySelector("#issueDate"),
  dueDate: document.querySelector("#dueDate"),
  issuedBody: document.querySelector("#issuedTableBody"),
  issuedEmpty: document.querySelector("#issuedEmpty"),
  issuedCount: document.querySelector("#issuedCount"),
  receiptModal: document.querySelector("#receiptModal")
};

const customSelectControllers = new Map();
const dialog = {
  backdrop: document.querySelector("#appDialog"),
  title: document.querySelector("#dialogTitle"),
  message: document.querySelector("#dialogMessage"),
  cancel: document.querySelector("#dialogCancel"),
  confirm: document.querySelector("#dialogConfirm")
}
```

```

};
let dialogResolver = null;

function readStorage(key, fallback) {
  try {
    const saved = JSON.parse(localStorage.getItem(key));
    return Array.isArray(saved) ? saved : fallback;
  } catch (error) {
    return fallback;
  }
}

function persistState() {
  localStorage.setItem(BOOKS_KEY, JSON.stringify(books));
  localStorage.setItem(ISSUED_KEY, JSON.stringify(issuedRecords));
}

function dateToInputValue(date) {
  const year = date.getFullYear();
  const month = String(date.getMonth() + 1).padStart(2, "0");
  const day = String(date.getDate()).padStart(2, "0");
  return `${year}-${month}-${day}`;
}

function parseDate(value) {
  const [year, month, day] = value.split("-").map(Number);
  return new Date(year, month - 1, day);
}

function formatDate(value) {
  if (!value) return "-";
  return parseDate(value).toLocaleDateString("en-IN", { day: "2-digit", month: "short", year: "numeric" });
}

function calculateDueDate(issueDate) {
  const dueDate = parseDate(issueDate);
  dueDate.setDate(dueDate.getDate() + LOAN_DAYS);
  return dateToInputValue(dueDate);
}

function getRecordStatus(record) {
  const today = new Date();
  today.setHours(0, 0, 0, 0);
  const dueDate = parseDate(record.dueDate);
  const overdueDays = Math.max(0, Math.floor((today - dueDate) / (1000 * 60 * 60 * 24)));
  return { overdueDays, fine: overdueDays * FINE_PER_DAY, isOverdue: overdueDays > 0 };
}

function escapeHtml(value) {
  return String(value).replace(/[\<>'"]/g, (character) => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", "'": "&#39;", '"': "&quot;" })[character]);
}

function enhanceSelect(select) {
  const wrapper = document.createElement("div");
  wrapper.className = "custom-select";
  select.parentNode.insertBefore(wrapper, select);
  wrapper.appendChild(select);
}

```

```

select.classList.add("native-select-hidden");

const trigger = document.createElement("button");
trigger.type = "button";
trigger.className = "custom-select-trigger";
trigger.setAttribute("aria-haspopup", "listbox");
trigger.setAttribute("aria-expanded", "false");
trigger.setAttribute("aria-controls", `${select.id}-menu`);

const label = document.createElement("span");
const chevron = document.createElement("span");
chevron.className = "select-chevron";
chevron.setAttribute("aria-hidden", "true");
trigger.append(label, chevron);

const menu = document.createElement("div");
menu.id = `${select.id}-menu`;
menu.className = "custom-select-menu";
menu.setAttribute("role", "listbox");
wrapper.append(trigger, menu);

const close = () => {
  wrapper.classList.remove("is-open");
  trigger.setAttribute("aria-expanded", "false");
};

const refresh = () => {
  const selectedOption = select.options[select.selectedIndex];
  label.textContent = selectedOption ? selectedOption.textContent : "Select an option";
  label.classList.toggle("select-placeholder", !select.value);
  menu.innerHTML = Array.from(select.options).map((option) => `<button class="custom-
select-option ${option.value ? "" : "is-placeholder"} ${option.value === select.value &&
option.value ? "is-selected" : ""}" type="button" role="option" aria-selected="${option.value
=== select.value}" data-
value="${escapeHtml(option.value)}">${escapeHtml(option.textContent)}</button>`).join("");
};

trigger.addEventListener("click", () => {
  const shouldOpen = !wrapper.classList.contains("is-open");
  document.querySelectorAll(".custom-select.is-open").forEach((openSelect) => {
    openSelect.classList.remove("is-open");
    openSelect.querySelector(".custom-select-trigger").setAttribute("aria-expanded",
"false");
  });
  wrapper.classList.toggle("is-open", shouldOpen);
  trigger.setAttribute("aria-expanded", String(shouldOpen));
});

trigger.addEventListener("keydown", (event) => {
  if (["Enter", " ", "ArrowDown"].includes(event.key)) {
    event.preventDefault();
    wrapper.classList.add("is-open");
    trigger.setAttribute("aria-expanded", "true");
    menu.querySelector(".custom-select-option").focus();
  }
  if (event.key === "Escape") close();
});

menu.addEventListener("click", (event) => {

```

```

        const option = event.target.closest(".custom-select-option");
        if (!option) return;
        select.value = option.dataset.value;
        select.dispatchEvent(new Event("change", { bubbles: true }));
        refresh();
        close();
        trigger.focus();
    });

    select.addEventListener("change", refresh);
    refresh();
    customSelectControllers.set(select, { refresh });
}

function showNotice(message, isError = false) {
    elements.notice.textContent = message;
    elements.notice.classList.toggle("error", isError);
    window.clearTimeout(showNotice.timer);
    showNotice.timer = window.setTimeout(() => {
        elements.notice.textContent = "";
        elements.notice.classList.remove("error");
    }, 4500);
}

function openDialog(message, { title = "Confirm action", confirmText = "Confirm", showCancel = true } = {}) {
    dialog.title.textContent = title;
    dialog.message.textContent = message;
    dialog.confirm.textContent = confirmText;
    dialog.cancel.hidden = !showCancel;
    dialog.backdrop.hidden = false;
    document.body.style.overflow = "hidden";
    dialog.confirm.focus();
    return new Promise((resolve) => { dialogResolver = resolve; });
}

function closeDialog(result) {
    dialog.backdrop.hidden = true;
    document.body.style.overflow = "";
    if (dialogResolver) {
        dialogResolver(result);
        dialogResolver = null;
    }
}

function renderDashboard() {
    const activeRecords = issuedRecords.filter((record) => !record.returnedAt);
    const overdueCount = activeRecords.filter((record) =>
getRecordStatus(record).isOverdue).length;
    const collectedFine = issuedRecords.reduce((sum, record) => sum +
Number(record.collectedFine || 0), 0);
    const totalVolumes = books.reduce((sum, book) => sum + Number(book.quantity), 0);
    document.querySelector("#totalBooks").textContent = totalVolumes;
    document.querySelector("#issuedBooks").textContent = activeRecords.length;
    document.querySelector("#overdueBooks").textContent = overdueCount;
    document.querySelector("#totalFine").textContent = `₹${collectedFine}`;
    elements.bookCount.textContent = `${books.length} title${books.length === 1 ? "" : "s"}`;
    elements.issuedCount.textContent = `${activeRecords.length} active
record${activeRecords.length === 1 ? "" : "s"}`;

```

```

}

function renderBooks() {
    const query = elements.bookSearch.value.trim().toLowerCase();
    const visibleBooks = books.filter((book) => `${book.title}
${book.code}`.toLowerCase().includes(query));
    elements.booksBody.innerHTML = visibleBooks.map((book) => `<tr>
<td><div class="book-
detail"><strong>${escapeHtml(book.title)}</strong><span>${escapeHtml(book.author)} ·
${escapeHtml(book.code)}</span></div></td>
<td>${escapeHtml(book.category)}</td>
<td><span class="stock ${book.quantity === 0 ? "empty" : ""}>${book.quantity}</span></td>
<td><div class="action-group"><button class="text-button edit" type="button" data-edit-
book="${escapeHtml(book.id)}">Edit</button><button class="text-button" type="button" data-
delete-book="${escapeHtml(book.id)}">Delete</button></div></td>
</tr>`).join("");
    elements.booksEmpty.hidden = visibleBooks.length !== 0;
}

function renderBookOptions() {
    const availableBooks = books.filter((book) => book.quantity > 0);
    elements.issueBook.innerHTML = `<option value="">Select a book with available
stock</option>${availableBooks.map((book) => `<option
value="${escapeHtml(book.id)}">${escapeHtml(book.title)} (${escapeHtml(book.code)}) -
${book.quantity} available</option>`).join("")}`;
    customSelectControllers.get(elements.issueBook)?.refresh();
}

function renderIssuedRecords() {
    const activeRecords = issuedRecords.filter((record) => !record.returnedAt);
    elements.issuedBody.innerHTML = activeRecords.map((record) => {
        const status = getRecordStatus(record);
        const book = books.find((item) => item.id === record.bookId);
        const title = book ? book.title : record.bookTitle;
        const statusText = status.isOverdue ? `Late / Overdue (₹${status.fine})` : "On Time";
        return `<tr>
<td><div class="student-
detail"><strong>${escapeHtml(record.studentName)}</strong><span>${escapeHtml(record.rollNumber)} ·
${escapeHtml(record.branchSemester)}</span></div></td>
<td><div class="book-
detail"><strong>${escapeHtml(title)}</strong><span>${escapeHtml(record.bookCode)}</span></div><
/td>
<td>${formatDate(record.issueDate)}</td><td>${formatDate(record.dueDate)}</td>
<td><span class="status-badge ${status.isOverdue ? "overdue" : "on-
time"}">${statusText}</span></td>
<td><div class="action-group"><button class="print-link" type="button" data-print-
record="${escapeHtml(record.id)}">Print slip</button><button class="return-button"
type="button" data-return-record="${escapeHtml(record.id)}">Return</button></div></td>
</tr>`;
    }).join("");
    elements.issuedEmpty.hidden = activeRecords.length !== 0;
}

function renderAll() {
    renderDashboard();
    renderBooks();
    renderBookOptions();
    renderIssuedRecords();
}

```

```

function resetIssueDates() {
  const today = dateToInputValue(new Date());
  elements.issueDate.value = today;
  elements.issueDate.max = today;
  elements.dueDate.value = calculateDueDate(today);
}

function resetBookForm() {
  editingBookId = null;
  elements.bookForm.reset();
  document.querySelector("#bookCategory").dispatchEvent(new Event("change"));
  elements.bookFormTitle.textContent = "Add a new book";
  elements.bookSubmitButton.innerHTML = '<span aria-hidden="true">+</span> Add book to library';
  elements.cancelBookEdit.hidden = true;
}

function editBook(bookId) {
  const book = books.find((item) => item.id === bookId);
  if (!book) return;
  editingBookId = bookId;
  elements.bookForm.elements.title.value = book.title;
  elements.bookForm.elements.author.value = book.author;
  elements.bookForm.elements.code.value = book.code;
  elements.bookForm.elements.category.value = book.category;
  elements.bookForm.elements.quantity.value = book.quantity;
  document.querySelector("#bookCategory").dispatchEvent(new Event("change"));
  elements.bookFormTitle.textContent = "Update book details";
  elements.bookSubmitButton.innerHTML = '<span aria-hidden="true">✓</span> Update book';
  elements.cancelBookEdit.hidden = false;
  elements.bookForm.scrollToView({ behavior: "smooth", block: "center" });
}

async function deleteBook(bookId) {
  const book = books.find((item) => item.id === bookId);
  if (!book) return;
  const hasActiveLoan = issuedRecords.some((record) => record.bookId === bookId && !record.returnedAt);
  if (hasActiveLoan) {
    showNotice("This book cannot be deleted while it is issued to a student.", true);
    return;
  }
  if (!await openDialog(`Delete "${book.title}" from the library catalogue?`, { title: "Delete book", confirmText: "Delete" })) return;
  books = books.filter((item) => item.id !== bookId);
  persistState();
  renderAll();
  showNotice("Book removed from the library catalogue.");
}

function addBook(event) {
  event.preventDefault();
  const formData = new FormData(elements.bookForm);
  const code = formData.get("code").trim().toUpperCase();
  if (books.some((book) => book.id !== editingBookId && book.code.toUpperCase() === code)) {
    showNotice("That accession / book code already exists. Use a unique code.", true);
    return;
  }
}

```

```

    const updatedBook = { id: editingBookId || `book-${Date.now()}`, title:
formData.get("title").trim(), author: formData.get("author").trim(), code, category:
formData.get("category"), quantity: Number(formData.get("quantity")) };
    const wasEditing = Boolean(editingBookId);
    if (wasEditing) {
        books = books.map((book) => book.id === editingBookId ? updatedBook : book);
    } else {
        books.push(updatedBook);
    }
    persistState();
    resetBookForm();
    renderAll();
    showNotice(wasEditing ? "Book details updated successfully." : "Book added to the library
catalogue.");
}

function issueBook(event) {
    event.preventDefault();
    const formData = new FormData(elements.issueForm);
    const book = books.find((item) => item.id === formData.get("bookId"));
    if (!book || book.quantity < 1) {
        showNotice("This book is no longer available. Please choose another title.", true);
        renderAll();
        return;
    }
    book.quantity -= 1;
    issuedRecords.push({ id: `issue-${Date.now()}`, bookId: book.id, bookTitle: book.title,
bookCode: book.code, studentName: formData.get("studentName").trim(), rollNumber:
formData.get("rollNumber").trim(), branchSemester: `${formData.get("branch")} -
${formData.get("semester")}`, issueDate: formData.get("issueDate"), dueDate:
formData.get("dueDate"), returnedAt: null, collectedFine: 0 });
    persistState();
    elements.issueForm.reset();
    resetIssueDates();
    renderAll();
    showNotice(`${book.title}` issued successfully. Return it by
${formatDate(formData.get("dueDate"))}.`);
}

async function returnBook(recordId) {
    const record = issuedRecords.find((item) => item.id === recordId);
    if (!record || record.returnedAt) return;
    const status = getRecordStatus(record);
    const fineMessage = status.fine > 0 ? ` A fine of ₹${status.fine} has been collected.` : "
No fine is due.";
    if (!await openDialog(`Record return for ${record.studentName}${fineMessage}`, { title:
"Confirm book return", confirmText: "Return book" })) return;
    const book = books.find((item) => item.id === record.bookId);
    if (book) book.quantity += 1;
    record.returnedAt = new Date().toISOString();
    record.collectedFine = status.fine;
    persistState();
    renderAll();
    await openDialog(`Book returned successfully.${fineMessage}`, { title: "Return recorded",
confirmText: "Done", showCancel: false });
}

function openReceipt(recordId) {
    const record = issuedRecords.find((item) => item.id === recordId);

```

```

    if (!record) return;
    const book = books.find((item) => item.id === record.bookId);
    document.querySelector("#receiptNumber").textContent = record.id.replace("issue-", "LIB-");
    document.querySelector("#receiptIssuedOn").textContent = formatDate(record.issueDate);
    document.querySelector("#receiptStudent").textContent = record.studentName;
    document.querySelector("#receiptRoll").textContent = record.rollNumber;
    document.querySelector("#receiptBranch").textContent = record.branchSemester;
    document.querySelector("#receiptCode").textContent = record.bookCode;
    document.querySelector("#receiptBook").textContent = book ? book.title : record.bookTitle;
    document.querySelector("#receiptIssueDate").textContent = formatDate(record.issueDate);
    document.querySelector("#receiptDueDate").textContent = formatDate(record.dueDate);
    elements.receiptModal.hidden = false;
    document.body.style.overflow = "hidden";
}

function closeReceipt() {
    elements.receiptModal.hidden = true;
    document.body.style.overflow = "";
}

elements.bookForm.addEventListener("submit", addBook);
elements.issueForm.addEventListener("submit", issueBook);
elements.bookSearch.addEventListener("input", renderBooks);
elements.issueDate.addEventListener("change", () => { elements.dueDate.value =
calculateDueDate(elements.issueDate.value); });
elements.booksBody.addEventListener("click", (event) => { const editButton =
event.target.closest("[data-edit-book]"); const deleteButton = event.target.closest("[data-
delete-book]"); if (editButton) editBook(editButton.dataset.editBook); if (deleteButton)
deleteBook(deleteButton.dataset.deleteBook); });
elements.cancelBookEdit.addEventListener("click", resetBookForm);
elements.issuedBody.addEventListener("click", (event) => { const returnButton =
event.target.closest("[data-return-record]"); const printButton = event.target.closest("[data-
print-record]"); if (returnButton) returnBook(returnButton.dataset.returnRecord); if
(printButton) openReceipt(printButton.dataset.printRecord); });
document.querySelector("#closeReceipt").addEventListener("click", closeReceipt);
document.querySelector("#printReceipt").addEventListener("click", () => window.print());
elements.receiptModal.addEventListener("click", (event) => { if (event.target ===
elements.receiptModal) closeReceipt(); });
document.addEventListener("keydown", (event) => { if (event.key === "Escape" &&
!elements.receiptModal.hidden) closeReceipt(); });
dialog.confirm.addEventListener("click", () => closeDialog(true));
dialog.cancel.addEventListener("click", () => closeDialog(false));
dialog.backdrop.addEventListener("click", (event) => { if (event.target === dialog.backdrop &&
!dialog.cancel.hidden) closeDialog(false); });
document.addEventListener("keydown", (event) => { if (event.key === "Escape" &&
!dialog.backdrop.hidden && !dialog.cancel.hidden) closeDialog(false); });
document.addEventListener("click", (event) => { if (!event.target.closest(".custom-select"))
document.querySelectorAll(".custom-select.is-open").forEach((select) => {
select.classList.remove("is-open"); select.querySelector(".custom-select-
trigger")?.setAttribute("aria-expanded", "false"); }); });

const today = new Date();
elements.todayLabel.textContent = today.toLocaleDateString("en-IN", { day: "2-digit", month:
"short", year: "numeric" });
elements.heroDate.textContent = today.toLocaleDateString("en-IN", { weekday: "long", day:
"numeric", month: "long", year: "numeric" });
[
    document.querySelector("#bookCategory"),
    document.querySelector("#studentBranch"),

```

```
document.querySelector("#studentSemester"),  
document.querySelector("#issueBook")  
].forEach(enhanceSelect);  
resetIssueDates();  
renderAll();
```

Prasant MMIT Amroha

6. CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The development of the Polytechnic Digital Library System as a Minor Project provided profound practical exposure to frontend web development. By deliberately avoiding heavy frameworks and databases, the project enforced a deep understanding of core JavaScript mechanics, DOM manipulation, responsive CSS layouts, and browser-based data storage.

The final product successfully meets the objectives of providing a lightweight, robust, and easy-to-use circulation desk portal. It automates critical library operations such as due date calculations and fine generation, significantly reducing the manual effort required by the departmental library staff.

6.2 Future Enhancements

While the current system functions efficiently as a client-side application, it has scope for massive scaling in future revisions (such as a Major Project). Future enhancements could include:

1. Backend Integration: Connecting the frontend to a Node.js or PHP backend to centralize data.
2. Relational Database: Migrating from LocalStorage to MySQL or MongoDB for secure, multi-device data access and preventing data loss on browser cache clears.
3. Authentication System: Implementing secure login systems for both Librarians (Admins) and Students (Users).
4. Notifications: Integrating SMS or Email APIs to send automated alerts to students 1-2 days before their book is due.
5. Advanced Analytics: Generating detailed PDF/CSV reports based on branch-wise book circulations and library usage statistics.